

Theorem 1. *Let $A \in \mathbb{C}^{n \times n}$, and let*

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

Then every complex polynomial p satisfies

$$\|p(A)\|_2 \leq 2 \max_{z \in W(A)} |p(z)|.$$

We write $\operatorname{Re} X = (X + X^*)/2$, and $\operatorname{alg}(T)$ for the unital algebra generated by a matrix T . The main point of the proof is the following positive-real completion lemma.

Lemma 1 (Positive-real completion). *Suppose that $T \in \mathbb{C}^{n \times n}$ has n distinct eigenvalues, all in the closed unit disk. Suppose that $H : \mathbb{D} \rightarrow \mathbb{C}^{n \times n}$ is analytic and has the following properties:*

$$\begin{aligned} H(0) &= I, & \operatorname{Re} H(z) &\succeq 0 \quad (z \in \mathbb{D}), \\ H(z) - (I - zT)^{-1} &\in \operatorname{alg}(T^*) \quad (z \in \mathbb{D}). \end{aligned}$$

Then $\|T\|_2 \leq 2$.

Proof. We first recall a consequence of the matrix Herglotz theorem. If $K : \mathbb{D} \rightarrow \mathbb{C}^{n \times n}$ is analytic and $\operatorname{Re} K \succeq 0$, then

$$\mathcal{L}_K(z, w) = \frac{K(z) + K(w)^*}{1 - z\bar{w}} \tag{1}$$

is a positive matrix-valued kernel. In other words, for arbitrary points $z_0, \dots, z_N \in \mathbb{D}$ and vectors $\xi_0, \dots, \xi_N \in \mathbb{C}^n$,

$$\sum_{i,j=0}^N \xi_i^* \mathcal{L}_K(z_i, z_j) \xi_j \geq 0. \tag{2}$$

For completeness, the matrix Herglotz representation is obtained by applying the scalar Herglotz representation to $x^*K(z)x$ and using polarization. It gives a positive matrix-valued measure M on the unit circle and a self-adjoint matrix C such that

$$K(z) = iC + \int_{|\zeta|=1} \frac{\zeta + z}{\zeta - z} dM(\zeta).$$

The elementary identity

$$\frac{1}{1 - z\bar{w}} \left(\frac{\zeta + z}{\zeta - z} + \overline{\frac{\zeta + w}{\zeta - w}} \right) = \frac{2}{(1 - z\bar{\zeta})(1 - \bar{w}\zeta)}$$

proves (2).

Choose a diagonalization

$$T = S\Lambda S^{-1}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad G = S^*S.$$

Every element of $\text{alg}(T^*)$ has the form $S^{-*}DS^*$, with D diagonal. The map $D \mapsto S^{-*}DS^*$ is a fixed linear isomorphism from the diagonal matrices onto $\text{alg}(T^*)$. Thus, when its inverse is applied to the analytic function $H(z) - (I - zT)^{-1}$, the resulting diagonal matrix $D(z)$ is analytic and satisfies $D(0) = 0$. Consequently, after putting $K(z) = S^*H(z)S$, the hypotheses give

$$K(z) = G(I - z\Lambda)^{-1} + D(z)G, \quad (3)$$

where $D(z) = \text{diag}(d_1(z), \dots, d_n(z))$. Congruence by S preserves positive real parts, so the kernel in (1), formed from this K , is positive.

Define three Hermitian matrices by

$$\begin{aligned} P_{ij} &= \frac{G_{ij}}{1 - \overline{\lambda_i}\lambda_j/4}, \\ R_{ij} &= \frac{G_{ij}}{1 - \overline{\lambda_i}\lambda_j/2}, \\ X &= R - P. \end{aligned} \quad (4)$$

Fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In (2), use the n points and vectors

$$z_i = \frac{\overline{\lambda_i}}{2}, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n),$$

and also use the point $z_0 = 0$ with the vector

$$\xi_0 = v = -G^{-1}Pu. \quad (5)$$

This choice removes the entire unknown term D in (3). Indeed, its contribution to the quadratic form in (2) is

$$2 \operatorname{Re} \sum_{i=1}^n \overline{u_i} d_i(z_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - z_i \overline{z_j}} \right).$$

The sum in parentheses is $(Gv + Pu)_i$, which is zero by (5).

It remains to calculate the contribution of $G(I - z\Lambda)^{-1}$. Between the points z_i, z_j it is

$$\frac{2G_{ij}}{(1 - \overline{\lambda_i}\lambda_j/2)(1 - \overline{\lambda_i}\lambda_j/4)} = (4R - 2P)_{ij}.$$

Between z_i and 0 it is $(G + R)_{ij}$, and at 0, 0 it is $2G$. Hence (2) and (5) give, for every u ,

$$\begin{aligned} 0 &\leq u^*(4R - 2P)u + 2\operatorname{Re}(u^*(G + R)v) + 2v^*Gv \\ &= u^*(4X - XG^{-1}P - PG^{-1}X)u. \end{aligned}$$

Thus

$$4X - XG^{-1}P - PG^{-1}X \succeq 0. \quad (6)$$

Set

$$\begin{aligned} C &= G^{1/2}\Lambda G^{-1/2}, \quad \hat{P} = G^{-1/2}PG^{-1/2}, \\ \hat{R} &= G^{-1/2}RG^{-1/2}, \quad \hat{X} = \hat{R} - \hat{P}. \end{aligned}$$

Expanding the two scalar kernels in (4) as geometric series gives

$$\hat{P} = \sum_{k=0}^{\infty} 4^{-k} C^{*k} C^k, \quad \hat{R} = \sum_{k=0}^{\infty} 2^{-k} C^{*k} C^k, \quad (7)$$

and therefore

$$\hat{X} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) C^{*k} C^k \succeq 0. \quad (8)$$

Indeed,

$$C^k = G^{1/2}\Lambda^k G^{-1/2},$$

so the powers C^k are uniformly bounded because $|\lambda_i| \leq 1$. The geometric weights therefore make both series converge in norm. Congruencing (6) by $G^{-1/2}$ yields

$$4\hat{X} - \hat{X}\hat{P} - \hat{P}\hat{X} \succeq 0. \quad (9)$$

Let e be an eigenvector of the Hermitian matrix $\hat{P}$, with $\hat{P}e = pe$. If $p > 2$, then (9) gives

$$0 \leq 2(2 - p)e^*\hat{X}e.$$

Since $\hat{X} \succeq 0$, this forces $\hat{X}e = 0$. The $k = 1$ summand in (8) is $C^*C/4$, so $Ce = 0$. Formula (7) then gives $\hat{P}e = e$, contradicting $p > 2$. It follows that

$$I \preceq \hat{P} \preceq 2I. \quad (10)$$

The first series in (7) satisfies the Stein identity

$$\hat{P} = I + \frac{1}{4}C^*\hat{P}C.$$

Using (10), we conclude that

$$C^*C \preceq C^*\widehat{P}C = 4(\widehat{P} - I) \preceq 4I.$$

Thus $\|C\|_2 \leq 2$. Finally, if $S = UG^{1/2}$ is the polar decomposition of S , then

$$T = UCU^*.$$

Therefore $\|T\|_2 \leq 2$, as claimed. $\square$

We next construct the positive-real completion from the numerical range.

Lemma 2 (The double-layer positive map). *Let $B \in \mathbb{C}^{n \times n}$, and let Ω be a bounded convex domain with continuously differentiable boundary such that $W(B) \subset \Omega$. If f is analytic on a neighborhood of $\overline{\Omega}$ and $\max_{\overline{\Omega}} |f| \leq 1$, put $T = f(B)$. If $\text{alg}(T) = \text{alg}(B)$, then there is an analytic function $H : \mathbb{D} \rightarrow \mathbb{C}^{n \times n}$ satisfying all the hypotheses of Lemma 1, except possibly the distinct-eigenvalue hypothesis on T .*

Proof. We also have

$$\sigma(B) \subset W(B) \subset \Omega.$$

Thus the spectral mapping theorem and the bound on f give

$$\sigma(T) = f(\sigma(B)) \subset \overline{\mathbb{D}}.$$

Orient $\partial\Omega$ counterclockwise. Let $\nu(\sigma)$ denote the outward unit normal and let ds denote arclength. Put

$$R_\sigma = (\sigma I - B)^{-1}$$

and define an operator-valued measure on $\partial\Omega$ by

$$d\mu(\sigma) = \frac{1}{2\pi} \left(\nu(\sigma)R_\sigma + \overline{\nu(\sigma)}R_\sigma^* \right) ds. \quad (11)$$

This measure is positive. Indeed,

$$\nu(\sigma)(\overline{\sigma}I - B^*) + \overline{\nu(\sigma)}(\sigma I - B) = 2\text{Re}(\overline{\nu(\sigma)}(\sigma I - B)) \succeq 0,$$

because the tangent line at σ supports the convex set $\overline{\Omega}$ and $W(B) \subset \Omega$. Multiplication on the left by R_σ^* and on the right by R_σ gives precisely the matrix in parentheses in (11).

Along the positively oriented boundary, $d\sigma = i\nu(\sigma)ds$. Cauchy's formula therefore gives

$$\frac{1}{2\pi} \int_{\partial\Omega} \nu(\sigma) R_\sigma d\sigma = I.$$

Taking adjoints deals with the second term in (11), and hence

$$\int_{\partial\Omega} d\mu = 2I. \quad (12)$$

Consequently,

$$\Phi(h) = \frac{1}{2} \int_{\partial\Omega} h(\sigma) d\mu(\sigma) \quad (13)$$

is a unital positive map from $C(\partial\Omega)$ into $\mathbb{C}^{n \times n}$.

For a function h analytic near $\overline{\Omega}$, define

$$g_h(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{h(\sigma)}}{\sigma - z} d\sigma, \quad z \in \Omega. \quad (14)$$

The first part of (11), Cauchy's formula, and the adjoint of (14) give

$$2\Phi(h) = h(B) + g_h(B)^*. \quad (15)$$

For $z \in \mathbb{D}$, consider the boundary function

$$h_z(\sigma) = \frac{1 + zf(\sigma)}{1 - zf(\sigma)}.$$

It has nonnegative real part. Define

$$H(z) = \Phi(h_z).$$

The map H is analytic and $H(0) = I$. Since the measure μ is Hermitian, $\Phi(\overline{h}) = \Phi(h)^*$; hence

$$\operatorname{Re} H(z) = \Phi(\operatorname{Re} h_z) \succeq 0.$$

Since

$$h_z = 1 + 2 \sum_{m=1}^{\infty} z^m f^m,$$

where the series converges uniformly on $\partial\Omega$ for each fixed $z \in \mathbb{D}$, the boundedness of Φ and (15) show that, with norm convergence,

$$H(z) = I + \sum_{m=1}^{\infty} z^m (T^m + g_{f^m}(B)^*). \quad (16)$$

It follows that

$$H(z) - (I - zT)^{-1} = \sum_{m=1}^{\infty} z^m g_{f^m}(B)^*. \quad (17)$$

Every $g_{f^m}(B)$ belongs to $\text{alg}(B)$, by the finite-dimensional holomorphic functional calculus (equivalently, by Hermite interpolation). The assumption $\text{alg}(T) = \text{alg}(B)$ therefore puts every partial sum on the right-hand side of (17) in $\text{alg}(T^*)$. The series converges in norm: the series in (16) does, as just noted, and $\sum_{m \geq 0} z^m T^m = (I - zT)^{-1}$ converges because $\rho(zT) < 1$. Finally, $\text{alg}(T^*)$ is finite-dimensional and hence norm-closed, so the limit in (17) also belongs to $\text{alg}(T^*)$. This proves the lemma. $\square$

Proof of the theorem. The numerical range is compact and convex. First choose a bounded convex domain Ω , with smooth boundary, such that

$$W(A) \subset \Omega.$$

We shall prove

$$\|p(A)\|_2 \leq 2 \max_{z \in \overline{\Omega}} |p(z)|. \quad (18)$$

Matrices with distinct eigenvalues are dense in $\mathbb{C}^{n \times n}$. Moreover,

$$W(B) \subset W(A) + \{z : |z| \leq \|B - A\|_2\}.$$

Thus A can be approximated by matrices B with distinct eigenvalues and with $W(B) \subset \Omega$. Since polynomial functional calculus is continuous, it is enough to establish (18) for each such B .

Fix one such B , and write its distinct eigenvalues as $\beta_1, \dots, \beta_n$. Put

$$M_\Omega = \max_{z \in \overline{\Omega}} |p(z)|.$$

If $M_\Omega = 0$, then p is identically zero and there is nothing to prove. Otherwise, put $f = p/M_\Omega$, and set

$$R_\Omega = \max_{z \in \overline{\Omega}} |z|.$$

For a complex number η close to zero, define

$$f_\eta(z) = \frac{f(z) + \eta z}{1 + |\eta|R_\Omega}. \quad (19)$$

Then $\max_{\overline{\Omega}} |f_\eta| \leq 1$. Apart from finitely many values of η , the numbers

$$f(\beta_i) + \eta\beta_i, \quad 1 \leq i \leq n,$$

are pairwise distinct. Choose a sequence of nonzero admissible values of η tending to zero. For each of them, the matrix

$$T_\eta = f_\eta(B)$$

has distinct eigenvalues. It also generates the same algebra as B . Indeed, $T_\eta \in \text{alg}(B)$, while Lagrange interpolation at the distinct numbers $f_\eta(\beta_i)$ gives a polynomial q such that $q(T_\eta) = B$. Hence

$$\text{alg}(T_\eta) = \text{alg}(B).$$

Apply Lemma 2 to B and f_η . The resulting positive-real function satisfies the hypotheses of Lemma 1; indeed, spectral mapping and $\max_{\overline{\Omega}} |f_\eta| \leq 1$ put every eigenvalue of T_η in the closed unit disk. Hence

$$\|f_\eta(B)\|_2 \leq 2.$$

Letting $\eta \rightarrow 0$ gives $\|f(B)\|_2 \leq 2$, or

$$\|p(B)\|_2 \leq 2M_\Omega.$$

Now let the simple-spectrum matrices B tend to A . This proves (18).

Finally, a compact convex set in the plane admits smooth bounded convex outer approximations converging to it in Hausdorff distance. Choose such domains Ω_k for $W(A)$. Uniform continuity of p on a fixed compact neighborhood of $W(A)$ gives

$$\max_{z \in \overline{\Omega}_k} |p(z)| \longrightarrow \max_{z \in W(A)} |p(z)|.$$

Applying (18) to Ω_k and taking the limit proves

$$\|p(A)\|_2 \leq 2 \max_{z \in W(A)} |p(z)|.$$

Since $W(A)$ is already compact, this is equivalently the assertion that its closure is a 2-spectral set for A . To spell out the standard equivalence, the spectrum of A lies in $W(A)$. If a rational function r has no pole on the compact convex set $W(A)$, choose $\delta > 0$ so that it has no pole on the closed δ -neighborhood of $W(A)$. That neighborhood is convex, so Runge approximation gives polynomials q_j converging uniformly to r there. Hence $q_j(A) \rightarrow r(A)$ by the holomorphic functional calculus, while $\max_{W(A)} |q_j| \rightarrow \max_{W(A)} |r|$. Applying the polynomial estimate to q_j and passing to the limit gives the same estimate for r , which is precisely the 2-spectral-set assertion. $\square$